The Positivity Wall: Where the Icosian Closure Object Sits in the RH Landscape

(VFD programme — expository companion)

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Abstract

This is an *expository* synthesis. We recall how the Riemann Hypothesis (RH) reduces to a single positivity, survey the many equivalent forms this positivity takes (spectral, algebraic, dynamical, geometric), and explain why the geometric routes surveyed here reach the same inequality without crossing it. We then state, as precisely as the literature allows, the single object whose construction would close it—a canonical Frobenius over $\mathbb{Z}$ whose degree-form is the Weil positivity, i.e. the arithmetic surface of the function-field analogue. **No new result on RH is claimed.** Every mathematical statement below is standard and credited; the contribution is organizational: a clean map of the boundary, with explicit stamps separating what is known, what is verified here, and what is interpretation. We then locate this bundle's own object on that map (§9): it reaches the shared floor and stops at the same wall, exactly as the survey predicts.

1 Scope

We claim no theorem about RH. The value of this companion is a single, honestly stamped picture: RH is one inequality; it appears in many costumes; geometry stages it but does not close it; and the missing step is a construction, not a search. It is the landscape *around* the object built in the companion papers (`the-closure-object`: the icosian object and its exact L-function $\zeta_K(s)\zeta_K(s-1)$, with ζ one factor but not isolated; `witness-from-triad`: a parameter-free Weil witness on the object's cuspidal face); §9 places those results here. Otherwise we only organize known mathematics.

2 RH as one positivity

By the Weil explicit-formula criterion [1], RH is equivalent to the nonnegativity of an explicit quadratic functional $W(f) \geq 0$ on a suitable space of test functions. Every route below is a re-expression of this one inequality.

3 One condition, many faces

The following are standard equivalent criteria and standard programme templates for RH. The first three are *proven equivalent* to RH; the last two are programme templates (one open over $\mathbb{Z}$, one a theorem in the function-field model). A single off-line zero falsifies the proven-equivalent criteria simultaneously:

- Weil positivity $W(f) \geq 0$ (spectral / Connes [2]) — equivalent to RH;

- the Li coefficients $\lambda_n \geq 0$ (algebraic [3]) — equivalent to RH;
- the de Bruijn–Newman constant $\Lambda \leq 0$ (dynamical; with $\Lambda \geq 0$ proven [5], so RH $\iff \Lambda = 0$);
- hyperbolicity of *all* Jensen polynomials of Ξ is equivalent to RH; GORZ [4] prove the fixed-degree/asymptotic part, not RH itself;
- the existence of a self-adjoint operator with the zeros as spectrum (Hilbert–Pólya) — a *programme*, not a proven equivalence;
- positivity of a Frobenius degree-form on a surface — a *theorem* in the function-field template [6, 7], with no built analogue over $\mathbb{Z}$.

Viewed as the proven-equivalent core, these are not many problems but one, seen from many sides; the last two are the geometric/spectral templates one hopes to transport.

4 The structure around the zeros (true, but not the wall)

The Riemann zeros are modelled, with strong numerical and partial-theorem support, by the statistics of a chaotic, time-reversal-broken (GUE, $\beta = 2$) spectrum: Montgomery [8] proves the pair correlation in a restricted range (conditionally), and Odlyzko [9] confirms it numerically to high precision; full GUE statistics remain conjectural. Summed as oscillations the zeros reproduce the primes with von-Mangoldt weights (the explicit formula, unconditional); and Rodgers–Tao give $\Lambda \geq 0$, so RH would force the de Bruijn–Newman constant to $\Lambda = 0$ (not known unconditionally). The explicit formula holds whether or not RH is true; the GUE statistics are a separate, largely empirical input. These constrain the shape of any solution; they do not supply the positivity.

5 The missing object

The function-field analogue of RH is a theorem precisely because the positivity is *geometric*: for a curve $C/\mathbb{F}_q$, the bound on Frobenius eigenvalues follows from the Hodge-index theorem on the surface $C \times C$. Over $\mathbb{Z}$ the analogous object—a canonical Frobenius whose degree-form is $W(f)$, living on a putative arithmetic surface $\operatorname{Spec}\mathbb{Z} \times_{\mathbb{F}_1} \operatorname{Spec}\mathbb{Z}$—is *unbuilt*. The leading programme towards it—Connes' trace-formula / noncommutative-geometry approach and the related $\mathbb{F}_1$ / absolute-geometry programmes (Connes–Consani [10])—recovers the explicit formula as a trace, but reduces RH to a *positivity of a trace pairing* that remains unproven. (Brave-new-ring heuristics model $\mathbb{F}_1$ by the sphere spectrum $\mathbb{S}$ and the surface by $\mathbb{Z} \otimes_{\mathbb{S}} \mathbb{Z}$; this is a separate suggestion, not part of Connes–Consani's site.) The positivity is the wall.

6 A table of symbols, and which one is a placeholder

The RH sentence can be written entirely in standard notation; the notation is *complete*. What is incomplete is the *referent* of one symbol: it names a hoped-for object that has no agreed construction. We make this explicit.

symbol	reading	status
$\mathrm{Re}(s) = \frac{1}{2}$	critical line	defined; computable
$\lvert\alpha_i\rvert = q^{1/2}$	Frobenius eigenvalues on a circle	*theorem* (function field)
$\xi(s) = \xi(1-s)$	the mirror / functional equation	defined; proven
$W(f) \geq 0$	Weil positivity	defined; *unproven* (= RH)
Frob_q	geometric Frobenius	a real morphism
$\mathrm{Spec}\,\mathbb{Z} \times_{\mathbb{F}_1} \mathrm{Spec}\,\mathbb{Z}$	arithmetic surface	**no construction carrying the required positivity**
$\mathbb{F}_1$	"field with one element"	**many frameworks; none supplies the RH object**

Several inequivalent $\mathbb{F}_1$-frameworks exist (§7); what none supplies is a canonical arithmetic surface carrying the required intersection theory and positivity. The reciprocal mirror $x \mapsto 1/x$ on $\mathbb{P}^1$ (radius $0 \leftrightarrow \infty$, the unit circle $|x| = 1$ fixed) is a schematic model of the involution behind $\xi(s) = \xi(1-s)$ under $s = \frac{1}{2} + it \leftrightarrow \log|x|$; that involution is built. The unbuilt entry is the $\mathbb{F}_1$-surface. Writing the symbol is legitimate; *constructing the object it should point to is exactly the open problem.*

7 A ledger of $\mathbb{F}_1$ candidates

Because $\mathbb{F}_1$ is the placeholder, it is natural to ask whether one may simply *define* it—"$\mathbb{F}_1 :=$ unity," say—and proceed. Many definitions exist; the question is never whether it can be *named* but whether the named object *carries* the explicit-formula positivity *without assuming it.* We classify the principal published candidates by three documented criteria: DEFINED (an agreed object exists), CARRIES (building the surface over it yields $W(f)$ as a genuine degree/intersection form), and HONEST (it does so without putting the conclusion in by hand). The verdicts below report the literature's status, not a computation: whether a given $\mathbb{F}_1$ proves RH *is* the open problem.

candidate	verdict	where it stops
$\mathbb{F}_1 = \{*\}$, trivial monoid ("unity")	INERT	no nontrivial Frobenius
$\mathbb{F}_1$-vector space = pointed set [11]	INERT	$q \to 1$ kills the arithmetic
Deitmar monoid schemes [12]	INERT	toric-like (finite-type) factors
Borger Λ-rings [13]	OPEN	supplies Frobenius lifts, not $W(f)$
Connes–Consani arithmetic site [10]	OPEN	trace recovered, positivity open
Deninger dynamical cohomology [14]	CONJECTURAL	cohomology hypothetical
"$\mathbb{F}_1$ defined so $\lvert\alpha\rvert = q^{1/2}$" (control)	CIRCULAR	assumes the conclusion

None of the listed candidates is known to supply the required positivity without an extra hypothesis. Every honestly-defined object is either INERT (real, but carries no zeros—the fate of "unity"), OPEN (Connes–Consani, Borger: reaches the wall, positivity not established), or CONJECTURAL (Deninger: the needed cohomology is hypothetical). The only candidate that *carries* the positivity outright is the CIRCULAR control, which assumes $|\alpha| = q^{1/2}$ and is rejected on the same provenance grounds that reject a least-squares fitted map. The honest summary: one can name a real object, or one that would cross; no listed framework is known to be both.

Scope of this ledger. This is not a classification of all possible absolute geometries. It records the status of representative *published* routes relevant to the explicit-formula positivity problem; $\mathbb{F}_1$ is a broad and active landscape, and the entries here are a snapshot, not a verdict on the field.

8 The joint void: a single door seen from two rooms (a conjectural picture)

This section is an interpretive picture, not a theorem. RH is approached from two sides that the Hilbert–Pólya philosophy and Connes' programme *hope* are one object seen twice: an *arithmetic* side (a Frobenius on the $\mathbb{F}_1$-surface whose degree-form would be $W(f) \geq 0$) and a *spectral* side (a self-adjoint $H = H^*$ whose spectrum would be the zeros—Hilbert–Pólya, Berry–Keating [15]). Tabulating what each side reaches today ("reach" = a partial construction exists; "part." = modelled but not fully built; "dark" = no construction):

requirement	arithmetic	spectral	
explicit/trace formula	reach	reach	shared floor (no RH)
global density (Weyl law)	reach	reach	
GUE local statistics (empirical)	part.	part.	
mirror $\xi(s) = \xi(1-s)$	reach	reach	
Frobenius / arithmetic surface	part.	*dark*	geometry's half (unbuilt)
self-adjoint H on a positive space	*dark*	part.	physics' half (no final H)
$W(f) \geq 0$ / (spec H = zeros, $H = H^*$)	*dark*	*dark*	**the door**

Two observations. First, the two sides *already meet* on the structure that does not require RH (the shared floor; GUE only empirically). Second, the programme's *hope* is that the arithmetic-dark and spectral-dark requirements are the same object missing its two faces. We emphasise the status carefully: Connes' analysis shows RH is equivalent to the positivity of a *trace pairing*; it does *not* prove that "$W(f) \geq 0$" and "$H = H^*$" are literally the same statement, and **no theorem identifies the arithmetic and spectral constructions.** What can be said honestly is that *both, under their own strong hypotheses, would yield RH*, and the programme conjectures a single underlying object. If that conjecture held, the "both" requirement would not widen the problem but pin the missing object toward one shape—a heuristic motivation, not a lever. It does not lower the wall: supplying the positivity by fiat on either face is circular, as above. The honest gain is only a sharper *picture*—"a single door, seen from two rooms"—with the floor on both sides and each doorpost, at most, half-built.

9 Where this bundle's object sits on the map

The companion papers place a concrete geometric object on the survey above, and it lands exactly where the survey predicts—on the shared floor, stopping at the same wall.

- **The closure object** (`the-closure-object`) realizes the Eisenstein L-function $\zeta_K(s)\,\zeta_K(s-1)$ *exactly* (every Euler factor verified from the maximal icosian lattice); the Riemann ζ is one of its four factors but is *not* isolated—a mechanically certified negative. This is a geometric route that *reaches* a real arithmetic object containing ζ without reaching ζ alone: it stages the arithmetic, it does not touch the location of the zeros.
- **The witness** (`witness-from-triad`) supplies, parameter-free from the object's cuspidal Brandt face, the *prime side* of the Weil functional $W(h)$ of §2—the very inequality this survey is about. It is calibrated on ζ (where the zeros are known) and verified $W(h) \geq 0$ on every test; the residual gap is exactly the universal positivity quantifier of §2, here *GRH for one cuspidal L-function*, not classical RH.

So the programme's object is not an exception to the map. It reaches the floor (it *builds* explicit-formula ingredients from geometry, with no fitting) and stops at the wall (the universal positivity quantifier), as every route in §§3–7 does. It is a genuine geometric instance of "stages the positivity, does not close it"—and, via the joint-void picture of the previous section, it supplies the arithmetic-side ingredients parameter-free while leaving the one dark line untouched.

10 Closing

Every route *surveyed here*—cavity, mirror, operator, loop, gas, flow, hyperbolicity, and the icosian closure object of this bundle—is a re-expression of one positivity, and re-expression does not prove it. The single missing step is the construction of an arithmetic surface and the positivity of its intersection form (equivalently, the universal positivity quantifier on a witness like the one this bundle builds). We make no claim of progress on that step; we have only drawn the boundary cleanly, placed this programme's object on it, and marked, at each face, exactly where known mathematics ends.

A note on interpretation. Several physically suggestive readings arise naturally when one draws these pictures (curvature, mirrors, resonance). They are *interpretation only*; no statement in this note depends on them, and none is claimed as mathematics.

References

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